

MINI WIND TURBINE MODEL

A Community Service Project report submitted in partial fulfillment of the requirements for

the award of

Bachelor of Technology

In

ELECTRONICS & COMMUNICATION ENGINEERING

By

M.DEEPTHI

(23BQ1A04A4)

M.DURGA BHAVANI

(23BQ1A0490)

N. MOHANA PRASANNA

(23BQ1A04B0)

M.DEEPTHI

(24BQ5A0410)

P. KEERTHI SAI

(23BQ1A04C3)

L.JYOTHI BAI

(23BQ1A0486)

N.BALA DURGA

(23BQ1A0496)



VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY

(AUTONOMOUS)

(Approved by AICTE and permanently affiliated to JNTUK)

Accredited by NBA and NAAC with 'A' Grade

NAMBUR (V), PEDAKAKANI (M), GUNTUR-522 508

November 2025

DECLARATION

We, the students of III B.Tech Electronics and Communication Engineering Department, Vasireddy Venkatadri Institute of Technology, hereby declare that we have completed the mandatory community service from 29th July 2025 to 2nd August 2025 in **Pedakakani** village under the guidance of **T.Vijaya Kumar**, Assistant Professor and submitted the report in partial fulfillment of the requirement for the award of Bachelor of Technology by the Jawaharlal Nehru Technological University, Kakinada (JNTUK) during the academic year 2025-2026.

STUDENT NAME	ROLL NUMBER	SIGNATURE
M. DEEPTHI	(23BQ1A04A4)	
P. KEERTHI SAI	(23BQ1A04C3)	
M. DURGA BHAVANI	(23BQ1A0490)	
L. JYOTHI BAI	(23BQ1A0486)	
N. MOHANA PRASANNA	(23BQ1A04B0)	
N. BALA DURGA	(23BQ1A0496)	
M. DEEPTHI	(24BQ5A0410)	

Endorsements

Faculty Guide/Coordinator

Head of the Department

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
VASIREDDY VENKATADRI INSTITUTE OF TECHNOLOGY: NAMBUR
(AUTONOMOUS)
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA



CERTIFICATE

This is to certify that **Ms. M.Deepthi(23BQ1A04A4), Ms. P.Keerthi Sai(23BQ1A04C3), Ms. M.Durga Bhavani(23BQ1A0490), Ms.L.Jyothi Bai(23BQ1A0486), Ms.N.Mohana Prasanna (23BQ1A04B0),Ms. N. Bala Durga (23BQ1A0496),Ms.M.Deepthi(23BQ5A0410)** of Vasireddy Venkatadri Institute of Technology underwent community service in Pedakakani Village from 29th July 2025 to 2nd August 2025 in partial fulfillment of the requirement of te degree for Bachelor of Technology in Electronics and Communication Engineering, JNTUK, during the academic year 2025-2026.

The overall performance of the Community Service Volunteer during his/her community service is found to be (Satisfactory/Good).

Mr.T.Vijaya Kumar
CSP Coordinator

Dr.M.Y. Bhanu Murthy
Head of the Department

ACKNOWLEDGEMENT

We express our sincere thanks wherever it is due

We express our sincere thanks to the Chairman of Vasireddy Venkatadri Institute of Technology, Sri Vasireddy Vidya Sagar, for providing us with a well-equipped infrastructure and an environment.

We thank Dr. Y. Mallikarjuna Reddy, Principal of Vasireddy Venkatadri Institute of Technology, Nambur, for providing us with the resources to carry out the project.

We express our sincere thanks to Dr. K. Giribabu, Dean of Academics, for providing support and a stimulating environment for developing the project.

Our sincere thanks to Prof. M. Y. Bhanu Murthy, Head of the Department, Department of ECE, for his cooperation and guidance, which helped us to make our project successful and complete in all aspects.

We also express our sincere thanks and are grateful to our CSP coordinator Mr.T.Vijaya Kumar, Assistant Professor, Department of ECE, for motivating us to make our project successful and fully complete. We are grateful for his precious guidance and suggestions.

We also place our floral gratitude to all other teaching staff and lab technicians for their constant support and advice throughout the project.

NAME OF THE CANDIDATES

M.DEEPTHI	(23BQ1A04A4)	P.KEERTHI	(23BQ1A04C3)
M.DURGA BHAVANI	(23BQ1A0490)	L.JYOTHI	(23BQ1A0486)
N.PRASANNA	(23BQ1A04B0)	N.DURGA	(23BQ1A0496)
M.DEEPTHI	(24BQ5A0410)		

TABLE OF CONTENTS

S.No	Description	Page No.
1	Chapter 1: Executive summary	1
2	Chapter 2: Socio-Economic Survey of Village	2
3	Chapter 3: Community Service Part	23
4	Chapter 4: Outcomes and Description	24
5	Chapter 5: Results and Conclusion	32

CHAPTER 1

EXECUTIVE SUMMARY

Community programs can expand the opportunities for youth to acquire personal and social assets and to experience the broad range of features of positive developmental settings. The activities I performed in the community service were Providing food to stray animals, Plantation in public parks, and cleaning up local parks. The activities are of great significance as they affected the surroundings a lot, it also developed a mindset of “cleaning our surroundings, helping animals, and planting more and more trees” in the minds of the people who acknowledged my work. The first task I performed was cleaning the local park. In day-to-day use, the local parks get dirty because of the kids and the people using them. The major problem of today's population is that they have become mean and self-centered, I noticed this problem when I noticed the mess, we have created in our surroundings (parks). The seats are dirty, and the park has more plastic wrappers than grasses. As I started cleaning the park the people started noticing me asked why am I wasting my time on cleaning the local park, they asked me put my efforts elsewhere but after a short conversation they got the motive of my efforts. They appreciated me as well as they knew the importance of keeping the surrounding clean. The second activity I performed was planting trees and flower plants. The local parkers suffer a lot of damage because of great amount of footfall of different age group people. Mostly the young kids play and destroys the plants in the parks. After lot of efforts even the Nagar Nigam is not able to manage all the parks and their plants. So, I took a small step my at least planting few plants in my arras local park for a change. I even taught the kids to take care of the plants rather destroying them so that they can to enjoy its beauty and benefits in the coming future. The third activity I did was feeding the stray dogs. After the Covid19 pandemic the population has been self-conscious and self-concentrated. Very less amount / percent of people is taking care of the stray animals. As the stray dogs are mostly dependent on the food we provide them, many dogs were starving when I tried to feed them. The ate a good amount of food, I got to know their situation so I decided to serve them some of their desired food daily for fewdays

Chapter 2

Socio-Economic Survey of Village

Taking details of Respondents

Day-1:- Date:31.07.2025

Respondent Name: Lakshamma

Education: 3rd Class

Occupation: Daily wage labour

Children: 2 boys studying in Government School

Family Income: ₹18,000 per month

Notable Issues in the Village: Frequent power cuts, poor road conditions

Photo with Respondent:



Respondent Name: Anjamma

Education: No formal education

Occupation: Homemaker

Children: 3 daughters (two married, one studying 9th class)

Family Income: ₹20,000 per month

Notable Issues in the Village: No government hospital nearby, no public transport facility

Photo with Respondent:



Respondent Name: Ramaswamy

Education: 7th Class

Occupation: Farmer

Children: 1 son working in city, 1 daughter studying Intermediate

Family Income: ₹30,000 per month

Notable Issues in the Village: Lack of proper irrigation facilities, low water supply

Photo with Respondent:



Respondent Name: Parvathamma

Education: 4th Class

Occupation: Selling vegetables in local market

Children: 1 boy doing B.Sc, 1 girl in 9th class

Family Income: ₹22,000 per month

Notable Issues in the Village: No proper drinking water, road damaged during rains

Photo with Respondent:



Respondent Name: Kamalamma

Education: 2nd Class

Occupation: Beedi rolling worker

Children: 1 son working in private company, 1 daughter married

Family Income: ₹15,000 per month

Notable Issues in the Village: No medical shop nearby, irregular bus services

Photo with Respondent:



Respondent Name: Jayamma

Education: 9th class

Occupation: Selling flowers near temple

Children: 1 daughter married

Family Income: ₹16,000 per month

Notable Issues in the Village: No market shed, no garbage cleaning facility

Photo with Respondent:



Respondent Name: Veerababu

Education: Intermediate

Occupation: Electric poles maintenance worker

Children: 1 son in 5th class

Family Income: ₹30,000 per month

Notable Issues in the Village: Street light issues, no medical emergency support

Photo with Respondent:



Day-2:- Date:01.08.2025

Respondent Name: Pushpavathi

Education: 8th Class

Occupation: Tailoring and embroidery work

Children: 2 girls studying in high school

Family Income: ₹24,000 per month

Notable Issues in the Village: No women self-help groups, water scarcity

Photo with Respondent:



Respondent Name: Anasuya

Education: 5th Class

Occupation: Housemaid

Children: 2 sons working in private firms

Family Income: ₹20,000 per month

Notable Issues in the Village: No PHC nearby, no proper bus shelter

Photo with Respondent:



Respondent Name: Govardhan

Education: 10th Class

Occupation: Mason

Children: 1 son married, 1 daughter working in textile shop

Family Income: ₹31,000 per month

Notable Issues in the Village: No proper housing scheme support, road damage

Photo with Respondent:



Respondent Name: Rukkamma

Education: 3rd Class

Occupation: Homemaker

Children: 2 daughters married

Family Income: ₹17,000 per month

Notable Issues in the Village: No proper sanitation, no waste bins in streets

Photo with Respondent:



Respondent Name: Chinnaiah

Education: 9th Class

Occupation: Blacksmith

Children: 2 sons studying ITI

Family Income: ₹25,000 per month

Notable Issues in the Village: Irregular electricity, no local spare parts shop

Photo with Respondent:



Respondent Name: Sulochana

Education: 4th Class

Occupation: Beedi worker

Children: 1 daughter married, 1 son working as mechanic

Family Income: ₹19,000 per month

Notable Issues in the Village: No medical shop, frequent power cuts

Photo with Respondent:



Respondent Name: Sharadamma
Education: 2nd Class
Occupation: Farm labourer
Children: 2 boys married
Family Income: ₹18,000 per month
Notable Issues in the Village: No public toilets, no street lights

Photo with Respondent:



Day-3:- Date:02.08.2025

Respondent Name: Rajeshwari

Education: 8th Class

Occupation: Small shop owner

Children: 1 son studying ITI

Family Income: ₹20,000 per month

Notable Issues: Power cuts, bad road conditions

Photo with Respondent:



Respondent Name: Parvathi
Education: 6th Class
Occupation: Beedi worker
Children: 1 son working, 1 daughter in school
Family Income: ₹17,000 per month
Notable Issues: Lack of toilets, poor health services

Photo with Respondent:



Respondent Name: Lalitha
Education: 9th Class
Occupation: Vegetable seller
Children: 2 sons working in nearby town
Family Income: ₹23,000 per month
Notable Issues: Unclean water, improper waste disposal

Photo with Respondent:



Respondent Name: Savitri
Education: 10th Class
Occupation: Anganwadi worker
Children: 1 son studying engineering
Family Income: ₹25,000 per month
Notable Issues: Irregular water supply, no nearby hospital

Photo with Respondent:



Respondent Name: Bhagyamma

Education: 5th Class

Occupation: Agricultural labourer

Children: 2 daughters (one married, one in 10th class)

Family Income: ₹19,000 per month

Notable Issues in the Village: No proper drainage, mosquitoes and sanitation problems

Photo with Respondent:



Respondent Name: Sumathi

Education: 10th Class

Occupation: ASHA worker

Children: 2 sons studying Intermediate

Family Income: ₹26,000 per month

Notable Issues in the Village: Lack of medical equipment in local clinic

Photo with Respondent:



Respondent Name: Padmamma
Education: 7th Class
Occupation: Selling flowers and coconuts near temple
Children: 1 daughter in 10th class
Family Income: ₹20,000 per month
Notable Issues in the Village: No garbage bins, poor sanitation

Photo with Respondent:



CHAPTER 3

COMMUNITY SERVICE PART

ACTIVITY LOG FOR THE FIRST WEEK

DAY & DATE	BRIEF DESCRIPTION OF THE DAILYACTIVITY	LEARNING OUTCOME	Person In-charge Signature
Day -1	We visited households to record issues regarding basic amenities like water, electricity, hygiene recording their responses.	Observed that there was a problem with unhygienic water and power cuts.	
Day -2	Surveyed and the farming fields and different equipment used for farming that work with electricity.	Learned that most of the farming equipment require electricity and frequent power cuts is a major problem.	
Day -3	Spoke with villagers to gather their perspective on problems: household water shortages, electricity shortage, drainage issues.	Evaluated how drainage issues are severe during rainy season, how power cuts are effecting their works.	
Day -4	Mapped local electricity infrastructure and interviewed residents about backup solutions (inverters, generators, solar panels).	Understood community reliance on costly or temporary backup systems and lack of awareness about renewable alternatives.	
Day -5	Studied existing renewable energy models (solar, wind, hybrid) and compared their cost-effectiveness and scalability.	Gained insight into how wind energy can complement solar power, especially during cloudy or nighttime conditions.	

CHAPTER 4

OUTCOMES DESCRIPTION

Details of the Socio-Economic Survey of the Village/Habitation (The questionnaire prepared for the survey in village and Village Resources) :

1. What are different plants and trees farming in the village?

Plants: maize, paddy, red gram, black gram, green gram, tomato, chilli, bitter gourd, brinjal
Trees: sapota and lemon are grown mostly in the village. we can see more number of sapota fruit tree farms across the road

2. What is the total count of the farming trees & plants in the village?

According to villagers point of view they cannot say proper number but the count varies from 10,000 - 13,000 trees and the plants varies from 30,000 to 35,000

3. Is Government water resources available for farming as well as drinking in the village?

No, government water resources available for farming and drinking in the village.
Everyone who cultivates farms have a separate bore well in their land and through

4. How many water tanks available in the village and what is the capacity of those tanks.

One tank available in the village

5. What are the common seasonal diseases village getting, mentions them season wise.

Common seasonal diseases in summer are fever, headache, vomiting, diarrhoea, sun stroke.
In winter and rainy, cough, cold and ever are observed. Dehydration problem occurs mostly in summer to the people as most of them are farmers while working in farms they have been get dehydrated

6. Which type of health facilities available or details about health issues commonly getting. (eg: Hospitals, Doctors available, RMP's, 108 & 104 services etc..)

Menno clinic hospital (American hospital) it is runned by anne yaarlagadda charitable trust. we are also having kauffman eye clinic and thompson dental clinic. the common health issues faced by the people are cough and cold during winter and rainy seasons. There is one Jagannna aarogya suraksha center, 1 RMP, 108 and 104 services are available.

7. Industrial facilities available/ factories available in the village premises?

8. The village doesn't have any factories or industrial units, which means its economy mainly depends on agriculture. Because of this, job options within the village are limited, and many residents may have to travel to nearby towns or cities for work.

9. .What are the Transportation resources available in the village? (Eg: Train, Bus and Other public transportation)

The village has access to train, auto, and limited bus transportation. Since there is no direct bus service within the village, residents must travel some distance towards the Tenali–Mangalagiri road to catch a bus. Auto services are available locally, with each auto accommodating up to 10 passengers at a time. Railway transport is also accessible through the Chiluvuru Railway Station, which is located on the Vijayawada–Gudur section of the Howrah–Chennai main line and operates under the Vijayawada Railway Division of the South Central Railway Zone. The station has two platforms and is classified as a Halt Grade-2 station based on earnings and passenger traffic.

10. Energy resources available in the village? (No.of Substations / Name of substrations)

one substation in chiluvuru (apspdcl-33/11kv sub station chiluvuru)

Financial resources like Banks or chitfund companies and other Govt. Schemesavailable.

Andhra bank chiluvuru branch later it is merged with union bank

11. Sanitization facilities in the village (panchayat/ Mandal/ Muncipaletc..)

This villages will come under the dugirala mandal

12. Agricultural facilities and their functionalities (Raithu Bharosa centres &Pesticide distribution centers etc.,)

One raithu barosa centre and one pesticide distribution centre are available

The problems we have identified in the community:

As part of our community visit, we explored a village located near our college. During the visit, we identified several issues such as poor electricity, mosquito infestation, inadequate security, lack of proper streetlights, more power cuts and incidents where residents are suffering especially in the night time due to lack of proper electricity. After reviewing all these concerns, our team decided to focus on developing a mini turbine model as our chosen solution.

Report of the mini-project

MINI WIND TURBINE MODEL

INTRODUCTION

In the present era, the global energy demand is increasing rapidly due to industrialization, population growth, and technological advancement. At the same time, the conventional energy sources such as coal, oil, and natural gas are depleting at an alarming rate and causing severe environmental problems like global warming and air pollution. Therefore, the focus has shifted towards renewable and eco-friendly energy sources that can meet future energy needs without harming the environment.

Among the various renewable sources, wind energy plays a significant role due to its abundance, sustainability, and pollution-free characteristics. Wind power is harnessed by wind turbines, which convert the kinetic energy of wind into mechanical and then into electrical energy. Large-scale wind turbines are used in wind farms to generate electricity for the grid, but the underlying working principle can be easily understood through a small-scale prototype model.

A Mini Wind Turbine Model is designed to demonstrate how electrical energy can be generated from the natural motion of air. It represents the basic working of real-world wind turbines in a compact and cost-effective form. The model includes components such as blades, a rotor, a generator (DC motor), and a base structure. When the wind strikes the blades, they rotate and initiate mechanical motion, which is converted into electrical energy by electromagnetic induction.

This generated energy can be utilized to operate low-power devices such as LEDs, and its magnitude can be observed using measurement systems to analyze the output performance.

The project also aims to spread awareness about renewable energy technologies among students and local communities. By developing this small-scale system, we can educate people about how simple mechanical setups can transform natural resources into electrical power, thereby encouraging sustainable practices and reducing dependency on non-renewable sources.

PROBLEM STATEMENT

The global reliance on non-renewable energy sources like coal and petroleum has led to serious issues such as air pollution, rising fuel prices, and energy shortages in rural and semi-urban areas. Many communities still suffer from irregular power supply, particularly during peak hours or adverse weather conditions. There is a pressing need for affordable, clean, and self-sufficient energy solutions that can be implemented even at a small scale.

To address this challenge, the proposed Mini Wind Turbine Model provides an eco-friendly solution that utilizes freely available wind energy. It demonstrates how power can be generated locally with minimal setup and cost. The system also helps learners understand the basic physics of energy conversion and its practical applications in real life.

OBJECTIVES

- To design and develop a compact model that efficiently converts wind energy into usable electrical energy.
- To demonstrate the working principle of wind energy conversion through a small-scale mechanical-to-electrical system.
- To study and analyze the relationship between wind speed, turbine speed, and electrical output.
- To measure and record output parameters such as voltage, current, and power under varying wind conditions.
- To compare the efficiency of different blade designs and materials in energy generation.
- To implement a rectifier and voltage regulator circuit for stable DC power output.
- To integrate sensors and a microcontroller (e.g., Arduino/NodeMCU) for monitoring real-time performance data.
- To develop a simple data acquisition system to log power output and environmental parameters.
- To explore the feasibility of powering small loads (LEDs, sensors, or mobile chargers) using the generated energy.
- To evaluate the performance of the turbine in indoor (fan-generated) and outdoor (natural wind) environments.
- To study the aerodynamics of turbine blades and their impact on rotational speed and efficiency.
- To design a cost-effective, portable, and eco-friendly renewable energy model suitable for educational use.

- To raise awareness about the importance of renewable and sustainable energy sources among students and communities.
- To provide hands-on experience in renewable energy system design, fabrication, and testing.
- To encourage further innovation and research in micro-scale renewable energy harvesting systems for rural applications.

METHODOLOGY

The development of the Mini Wind Turbine Model involves several stages, from conceptual design to functional testing. The step-by-step methodology is as follows:

1. Conceptual Design

A basic design is sketched that includes blades, rotor, shaft, and generator arrangement. The design ensures that the model efficiently captures the wind energy and converts it into electrical energy. The number of blades and their angles are optimized for better rotation with minimal resistance.

2. Component Selection

Lightweight materials such as plastic or PVC are used for blades to achieve smooth rotation. The rotor is attached to a small DC motor that acts as the generator. Supporting structures are made from strong, stable materials like wood or acrylic for firm mounting.

3. Mechanical Assembly

The blades are symmetrically attached to the rotor hub and balanced to prevent vibration. The rotor shaft is aligned with the DC motor shaft for efficient mechanical coupling. The entire assembly is fixed on a stable base that allows free rotation when air or wind is applied.

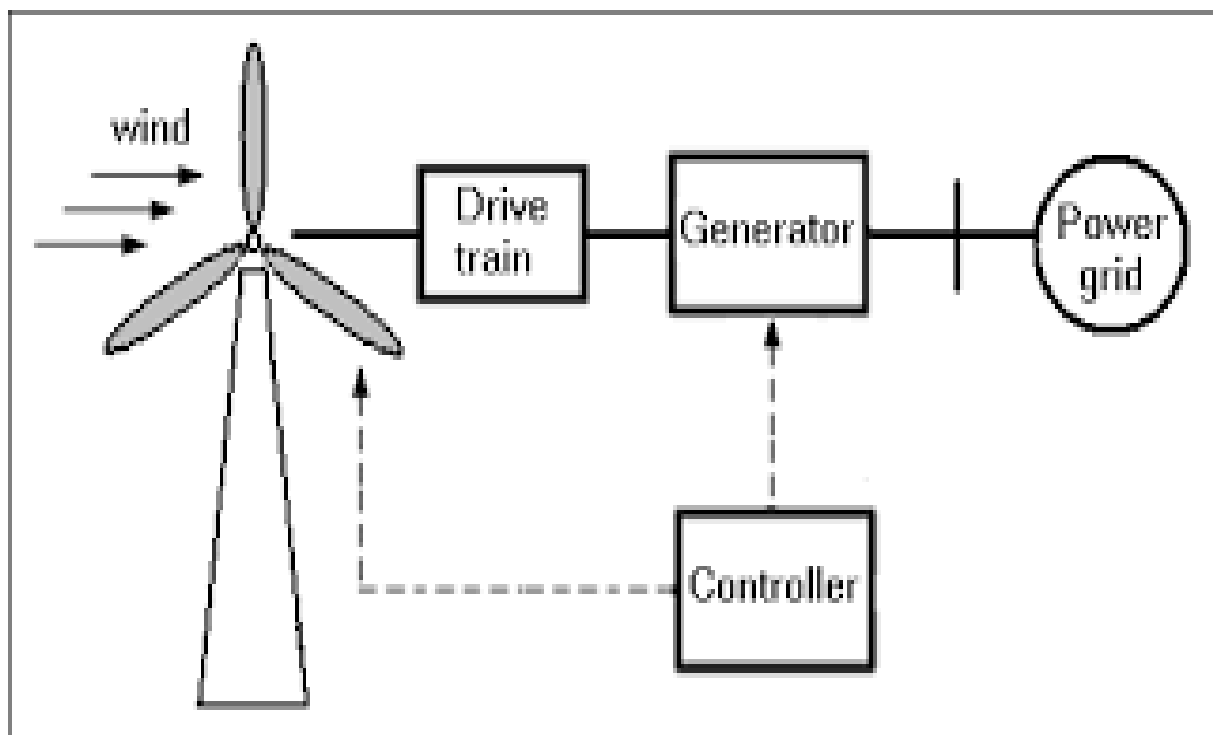
4. Electrical Connection

The generated electrical output from the DC motor is connected to a small load (like LEDs) through proper wiring. A measurement setup is included to observe output voltage and current, which helps determine the efficiency and output power of the system.

5. Testing and Observation

A table fan or controlled airflow is used to simulate wind at different speeds. For each condition, the output readings are recorded and compared. The relationship between wind speed, blade rotation, and generated voltage is analyzed to evaluate performance.

BLOCK DIAGRAM



This block diagram represents a Wind Energy Conversion System (WECS), which converts the kinetic energy of wind into electrical energy that can be supplied to the power grid. The main components involved in this process are the wind turbine, drive train, generator, controller, and power grid.

The wind turbine is the first component in the system. It captures the kinetic energy present in the moving air. When the wind blows, it causes the blades of the turbine to rotate. The amount of energy captured depends on the wind speed and the design of the blades. The rotating blades convert the wind's kinetic energy into mechanical rotational energy.

The drive train is connected to the turbine and is responsible for transferring the mechanical energy from the turbine rotor to the generator. It typically consists of a shaft and a gearbox. The gearbox increases the low-speed rotation from the turbine into a high-speed rotation suitable for the generator. This ensures efficient energy transfer from the turbine to the generator.

The generator converts the mechanical energy coming from the drive train into electrical energy. When the high-speed shaft rotates the generator's rotor, an electric current is induced, and alternating current (AC) power is produced. This electrical output can then be used for power distribution after synchronization with the grid.

The controller plays an essential role in regulating the operation of the entire system. It monitors various parameters like wind speed, rotor speed, and generator output. The controller adjusts the blade pitch angle and turbine speed to achieve maximum efficiency and protect the turbine from damage during high wind speeds. It also ensures that the generated power matches the voltage and frequency requirements of the grid. Finally, the generated electricity is supplied to the power grid, where it is distributed to consumers. Before integration into the grid, the output is synchronized to match grid standards.

In summary, the block diagram illustrates the overall working of a wind energy system — where wind energy is converted into mechanical energy by the turbine, then into electrical energy by the generator, and finally delivered to the power grid under the supervision of a controller for efficient and stable operation. In conclusion, the block diagram clearly shows how wind energy is converted into electrical energy through a systematic process involving the turbine, drive train, generator, and controller. The turbine captures wind energy, which is then transformed into mechanical and finally electrical energy. The controller ensures efficient, safe, and stable power generation before supplying it to the power grid.

DESIGN DESCRIPTION

The Mini Wind Turbine Model is built to demonstrate the energy conversion process from wind to electricity. The design operates on the principle of electromagnetic induction, where the mechanical rotation of a conductor in a magnetic field induces an electromotive force (EMF). The generated voltage depends on the speed of rotation and the magnetic strength inside the generator.

When airflow strikes the turbine blades, they capture the kinetic energy of wind and begin to rotate. This mechanical energy is transferred to the rotor shaft and subsequently to the DC motor. The motor acts as a generator by converting mechanical energy into electrical energy. The generated voltage is then transferred through connecting wires to a load, typically small LED lamps, which serve as visual indicators of energy generation.

A separate measurement arrangement is integrated to calculate the generated voltage and corresponding power output. By varying the wind speed using a fan, students can observe how output voltage and brightness change, indicating real-time variation in power generation. This demonstrates how larger turbines operate under natural wind variations in real-world conditions.

The model uses three to six blades, designed with lightweight material and a suitable pitch angle (10–15 degrees). The angle helps maximize wind capture efficiency and ensures a steady rotational motion even at low wind speeds. The setup is placed on a strong base to maintain balance during high airflow.

WORKING PRINCIPLE

The working principle of the Mini Wind Turbine Model can be explained in three stages: energy capture, energy conversion, and energy utilization. In the first stage, *energy capture*, the wind energy present in the atmosphere strikes the blades of the turbine, causing them to spin. The rotational kinetic energy generated depends on the wind velocity and the aerodynamic design of the blades. In the second stage, energy conversion, the rotor shaft connected to the blades transfers the mechanical energy to a DC motor, which acts as a generator. As the motor's internal magnetic field interacts with the rotating coil, an electromotive force (EMF) is induced according to Faraday's law of electromagnetic induction, converting mechanical energy into electrical energy. In the final stage, energy utilization, the generated electrical energy flows through connecting wires to a load such as LEDs. The brightness of the LEDs indicates the output power, and as the wind speed increases, both the voltage and current rise, resulting in greater illumination. Thus, the mini wind turbine efficiently demonstrates the conversion of wind energy into usable electrical energy.

CHAPTER 5

RESULTS AND CONCLUSIONS

RESULTS AND DISCUSSION

The Mini Wind Turbine Model successfully demonstrated the fundamental principle of converting wind energy into electrical energy through a compact and cost-effective setup. The model operated efficiently during testing, validating its ability to generate measurable voltage and current under different wind conditions. It effectively showcased how renewable energy can be utilized to produce electricity without relying on conventional fossil fuel-based sources. The model's performance indicated that even at low wind speeds, a small yet continuous power output could be maintained, which is suitable for lighting small loads such as LEDs or charging miniature devices. This result supports the concept that small-scale wind systems can provide meaningful contributions to decentralized energy generation, especially in rural and remote regions.

A major finding of this project was its practical applicability in addressing frequent power cuts. Many households in the surveyed area experience unpredictable and extended power outages, especially in regions with underdeveloped infrastructure. The mini wind turbine served as a demonstration of how communities can adopt small, affordable renewable systems to achieve partial energy independence. Although the power generated by the prototype is modest, it effectively symbolizes how renewable energy can bridge gaps in electricity supply. This highlights the model's potential to support low-power applications such as street lighting, charging mobile devices, or providing emergency lighting during blackouts.

The model also proved successful in promoting sustainable energy awareness. During demonstrations and surveys, local participants expressed increased interest in renewable energy after understanding how the turbine converts natural wind into usable power. The educational value of the model extends beyond its technical performance—it serves as an excellent teaching tool to illustrate basic energy conversion principles to students and the community. Its simplicity helps people visualize the real-world application of physics and electrical engineering concepts such as Faraday's law of electromagnetic induction, kinetic-to-electrical energy transformation, and mechanical power transmission.

Furthermore, the model's design and construction emphasized the importance of low-cost innovation. Using easily available materials like lightweight blades, a small DC motor, and recycled support structures, the model maintained both efficiency and affordability. This makes it replicable in rural or semi-urban areas where budget constraints often prevent investment in large-scale renewable infrastructure. The design also demonstrates scalability—small design modifications, such as using a larger motor or aerodynamic blades, can significantly increase the power output. Thus, the project effectively merges educational, environmental,

and practical objectives.

Another key observation was related to the relationship between wind speed and power generation. The output voltage and current increased proportionally with wind velocity, confirming theoretical predictions. The LEDs connected to the system glowed brighter as wind speed rose, indicating higher power generation. These real-time observations provided valuable insights into performance characteristics and system efficiency. Although power fluctuations were noticed at irregular wind speeds, the consistency of the output under moderate wind conditions reflected the stability of the model. Future improvements may include integrating a voltage regulator or battery storage unit to stabilize the output and store excess energy.

Overall, the results emphasize that even a small-scale wind turbine model can contribute significantly to energy awareness and rural electrification strategies. It not only highlights the feasibility of renewable energy adoption but also motivates individuals to explore and experiment with sustainable energy solutions. The success of the project lies not only in power generation but also in its impact on understanding and promoting eco-friendly technologies among students and local communities.

CONCLUSION

In conclusion, the Mini Wind Turbine Project demonstrates that wind energy can be effectively harnessed on a small scale to produce usable electrical energy through a simple and affordable setup. The project successfully addresses the issue of frequent power cuts by offering an innovative, sustainable alternative to conventional energy sources. By transforming the kinetic energy of wind into electrical energy using basic materials, the model showcases how renewable energy can be accessible to everyone, regardless of economic constraints. It serves as both a practical and educational tool that encourages experimentation and innovation in renewable energy utilization.

Through surveys, prototype development, and testing, the project validated that wind energy is not only an abundant natural resource but also a viable means to meet small-scale power demands. The implementation of this model demonstrated that clean energy solutions can be developed using locally available materials without the need for expensive technologies. This approach promotes self-reliance in energy generation, particularly in rural and semi-urban regions where power supply remains inconsistent. By engaging communities in renewable energy projects, the initiative fosters awareness, skill development, and environmental responsibility.

The project also contributes significantly to the promotion of sustainable development goals (SDGs), particularly those related to affordable and clean energy, innovation, and climate action. It serves as an inspiring example for students, educators, and local innovators to pursue clean technology solutions. Beyond its immediate technical application, the project's broader impact lies in building an understanding of how

small, localized energy systems can collectively support global sustainability efforts. It empowers individuals and communities to take part in the transition toward green energy and reduced carbon emissions.

Moreover, the mini wind turbine model opens avenues for further research and enhancement. Future developments may include integrating energy storage systems, employing advanced aerodynamic blades, or coupling with solar panels to form hybrid renewable systems. Such improvements can make the design more efficient and capable of supporting larger electrical loads. The continued evolution of this project can transform it from an educational prototype into a practical micro-generation system for rural electrification.

Ultimately, this project demonstrates that the path to sustainable energy begins with small, innovative steps. By merging science, creativity, and social responsibility, the Mini Wind Turbine Model provides a meaningful contribution toward building a cleaner and more resilient energy future. It not only generates power but also ignites curiosity, inspires change, and empowers communities to embrace renewable energy as a key solution to modern energy challenges.

Student Self-Evaluation for the Community Service Project

Student Name:

Registration No:

Period of CSP: From: To:

Date of Evaluation:

Name of the Person in-charge:

Address with mobile number:

Please rate your performance in the following areas:

Rating Scale: 1 is lowest and 5 is highest rank

1) Oral communication	1	2	3	4	5
2) Written communication	1	2	3	4	5
3) Proactiveness	1	2	3	4	5
4) Interaction ability with community	1	2	3	4	5
5) Positive Attitude	1	2	3	4	5
6) Self-confidence	1	2	3	4	5
7) Ability to learn	1	2	3	4	5
8) Work Plan and organization	1	2	3	4	5
9) Professionalism	1	2	3	4	5
10) Creativity	1	2	3	4	5
11) Quality of work done	1	2	3	4	5
12) Time Management	1	2	3	4	5
13) Understanding the Community	1	2	3	4	5
14) Achievement of Desired Outcomes	1	2	3	4	5
15) OVERALL PERFORMANCE	1	2	3	4	5

Date:

Signature of the Student

Evaluation by the Person in-charge in the Community/Habitation

Student Name:

Registration No:

Period of CSP: From: To:

Date of Evaluation:

Name of the Person in-charge:

Address with mobile number:

Please rate the student's performance in the following areas:

Please note that your evaluation shall be done independent of the student's self-evaluation

Rating Scale: 1 is lowest and 5 is highest rank

1) Oral communication	1	2	3	4	5
2) Written communication	1	2	3	4	5
3) Proactiveness	1	2	3	4	5
4) Interaction ability with community	1	2	3	4	5
5) Positive Attitude	1	2	3	4	5
6) Self-confidence	1	2	3	4	5
7) Ability to learn	1	2	3	4	5
8) Work Plan and organization	1	2	3	4	5
9) Professionalism	1	2	3	4	5
10) Creativity	1	2	3	4	5
11) Quality of work done	1	2	3	4	5
12) Time Management	1	2	3	4	5
13) Understanding the Community	1	2	3	4	5
14) Achievement of Desired Outcomes	1	2	3	4	5
15) OVERALL PERFORMANCE	1	2	3	4	5

Date:

Signature of the Supervisor